

UST'-ISHIM, Russian Federation

WMO: 283820

Lat: 57.683N

Lon: 71.183E

Elev: 183

StdP: 14.60

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.7	-28.5	-39.3	0.6	-33.5	-34.1	0.8	-28.3	17.8	14.1	16.0	15.7	2.1	280	0.646

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.9	83.6	68.3	80.7	66.6	77.5	64.8	70.5	80.0	68.7	77.3	66.7	74.7	7.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.2	100.8	75.7	65.4	94.4	73.2	63.6	88.4	71.1	34.6	79.7	33.0	77.4	31.4	74.8	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.7	16.5	14.5		-39.0	88.9	6.7	2.9	-43.9	91.0	-47.8	92.7	-51.6	94.3	-56.5	96.4	
				-39.4	73.7	6.6	2.6	-44.1	75.6	-48.0	77.1	-51.7	78.6	-56.5	80.5	
				DB												
				WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	33.4	-2.2	3.1	19.6	36.0	50.9	61.8	64.4	59.5	48.7	36.0	17.4	4.2	(6)
(7)		DBStd	25.75	15.10	13.96	11.03	10.03	9.29	8.37	7.00	6.86	7.72	8.90	14.33	15.65	(7)
(8)		HDD50	7379	1619	1314	942	428	107	11	1	6	117	438	977	1419	(8)
(9)		HDD65	11697	2084	1734	1407	871	445	157	100	198	490	900	1427	1884	(9)
(10)		CDD50	1337	0	0	0	8	135	364	448	300	78	4	0	0	(10)
(11)		CDD65	176	0	0	0	0	7	59	82	27	1	0	0	0	(11)
(12)		CDH74	1539	0	0	0	2	146	526	634	214	17	0	0	0	(12)
(13)		CDH80	324	0	0	0	0	33	121	132	38	0	0	0	0	(13)
(14)	Wind	WSAvg	6.5	5.3	5.6	7.1	7.6	7.6	6.5	6.1	6.1	6.4	6.9	6.9	6.1	(14)
(15)	Precipitation	PrecAvg	19.3	0.9	0.7	0.9	1.1	1.7	2.3	2.9	2.7	2.0	1.6	1.5	1.2	(15)
(16)		PrecMax	28.5	1.6	1.5	2.2	2.2	4.0	5.6	8.7	5.7	3.5	4.1	2.8	2.4	(16)
(17)		PrecMin	12.1	0.1	0.0	0.0	0.0	0.3	0.4	0.6	0.9	0.6	0.1	0.3	0.4	(17)
(18)		PrecStd	3.4	0.5	0.3	0.5	0.6	0.7	1.1	1.9	1.1	0.9	0.9	0.6	0.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.4	36.3	45.4	70.1	83.9	87.4	87.3	84.7	76.4	62.6	43.0	35.3	(19)
MCWB			32.0	32.7	37.4	53.1	61.1	67.3	71.1	69.8	60.7	51.4	39.9	34.2	(20)	
2%		DB	28.7	31.7	40.2	62.6	77.7	83.6	83.3	79.3	70.2	56.5	39.4	32.5	(21)	
		MCWB	27.4	29.8	34.5	50.1	58.8	67.7	69.5	66.5	58.7	48.5	37.3	31.6	(22)	
5%		DB	25.0	28.2	37.2	56.3	73.2	80.1	80.8	75.8	65.2	52.2	36.4	28.8	(23)	
		MCWB	23.9	26.6	32.9	46.0	57.2	66.3	67.8	65.2	56.3	46.4	34.6	27.5	(24)	
10%		DB	20.5	23.9	34.7	51.6	68.1	76.5	77.4	71.9	61.2	48.6	33.8	25.3	(25)	
		MCWB	19.4	22.1	31.4	43.4	54.6	64.4	66.2	63.2	54.1	44.2	32.6	24.1	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.3	33.3	38.2	54.3	63.3	72.8	73.2	71.1	62.8	52.8	40.7	34.4	(27)
MADB			33.4	35.5	43.6	67.5	78.7	81.4	83.5	81.7	71.2	59.8	42.7	34.9	(28)	
2%		WB	27.7	29.8	35.4	50.5	60.8	69.8	71.1	68.4	59.9	49.6	37.7	31.5	(29)	
		MADB	28.8	31.6	39.0	60.7	75.0	79.7	80.7	76.5	68.2	54.7	39.3	32.4	(30)	
5%		WB	23.8	26.5	33.7	47.2	58.3	67.6	69.3	66.3	57.3	47.1	34.7	27.8	(31)	
		MADB	24.8	28.1	36.7	55.4	70.3	77.0	78.2	73.4	63.8	51.2	36.0	28.8	(32)	
10%		WB	19.6	22.2	31.6	43.6	55.7	65.5	67.5	64.1	54.8	44.5	32.8	24.2	(33)	
		MADB	20.5	23.8	34.5	50.8	65.7	74.3	75.3	70.7	60.2	48.7	33.8	25.3	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	15.4	19.5	19.7	18.3	20.7	18.4	16.9	16.1	15.2	11.8	11.1	14.2	(35)
MADB			13.8	15.1	18.2	26.2	28.7	24.2	21.9	21.9	22.3	19.1	8.7	11.5	(36)	
MCWBR			13.6	13.6	13.7	15.5	12.9	10.8	10.0	10.7	11.6	12.3	7.9	11.1	(37)	
5% WB		MADB	14.1	15.1	15.5	24.5	25.5	21.3	19.6	19.8	20.1	17.7	8.6	11.2	(38)	
		MCWBR	13.8	13.9	12.2	15.5	13.1	10.7	9.9	10.5	11.6	12.2	8.2	11.1	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.275	0.311	0.369	0.408	0.403	0.409	0.427	0.423	0.375	0.349	0.306	0.275	(40)	
(41)		taud	2.021	2.015	1.929	2.050	2.252	2.304	2.251	2.232	2.324	2.243	2.086	1.969	(41)	
(42)		Ebn at Noon	172	220	236	251	263	263	254	246	240	208	157	128	(42)	
(43)		Edh at Noon	18	30	43	45	39	38	39	37	29	24	17	15	(43)	
(44)	All-Sky Solar Radiation	RadAvg	215	528	991	1411	1643	1757	1689	1278	836	416	187	125	(44)	
(45)		RadStd	19	41	69	118	128	137	167	107	87	54	35	16	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page